Cover letter

Journal of Number Theory 25 October 2026

Unsent submission correspondence; prepared for author review.

Dear Editors,

Please consider the manuscript “The optimal log log q bound in the Pólya-Vinogradov inequality” by Deep Bhattacharjee and Ushashi Bhattacharya.

The manuscript derives the optimal-order log log q character-sum bound and related logarithmic-derivative results from the credited zero-free theorem, with explicit parameters and a smooth Mobius equivalence. The complete formal source and exact endpoint audit passed. OpenAI’s zero-free proof remains credited; no newly independent proof is claimed.

The character-sum, Dirichlet L-function and logarithmic-derivative results fit analytic number theory and the journal’s detailed technical-article scope.

The prior public preprint is `https://hexagonmath.org/2610.00256v2`. The research manuscript/source archive is `https://doi.org/10.5281/zenodo.23629139`. Editable sources and verification records are supplied with the manuscript.

The complete current source build, all 19 original and seven closed endpoint axiom reports, and all 20 analytic component reports passed. The reports use only propext, Classical.choice and Quot.sound. The separate written cyclotomic identification is not a new Lean statement, and the revised exposition is not a line-by-line formalisation. No independent zero-free proof is claimed.

The manuscript and the accompanying declaration disclose the substantive use of Claude and OpenAI tools. The submitting author has declared that all parts have been checked and understood by a human. The named human authors retain responsibility for the content.

Sincerely,

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